Dynamic Systems Development Method Practices

Rohayanti Hassan
Faculty of Computing
Universiti Teknologi Malaysia

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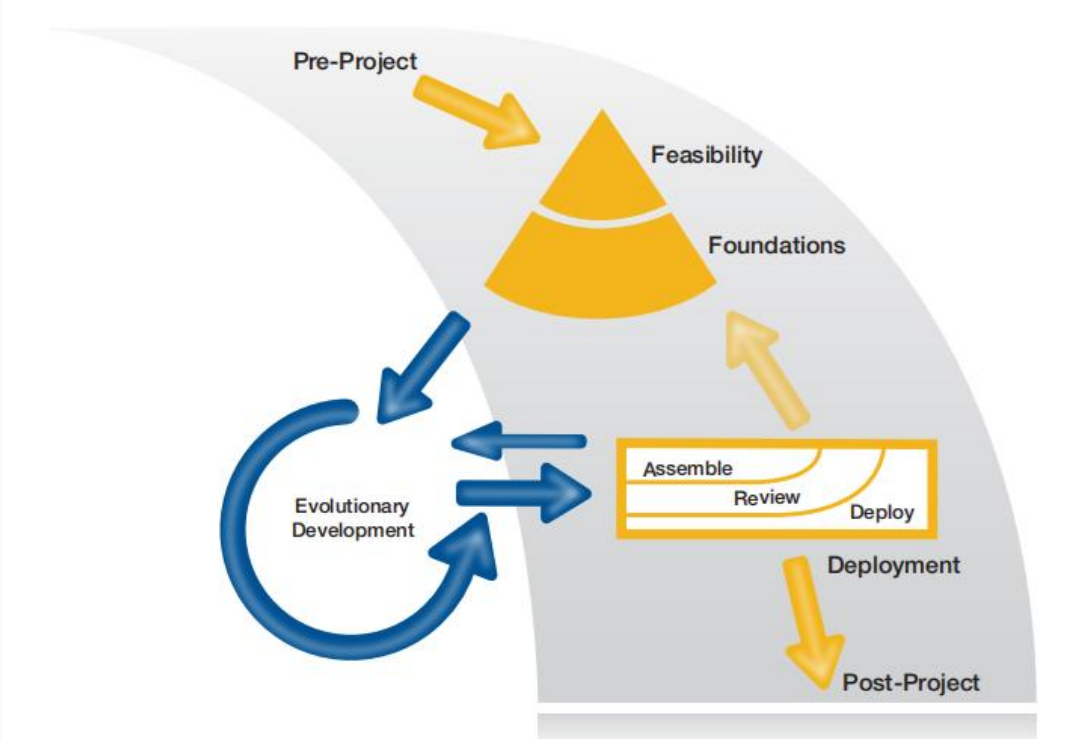
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Outline

- Overview
- Pre - project Phase
- Feasibility Phase
- Foundation Phase
- Evolutionary Development Phase
- Deployment Phase
- Post - project Phase
- The Lifecycle in practice
- Configuring DSDM for Scalability and Formality

Overview : The DSDM Process

- Process is used to create lifecycle for each project
- Each phase has defined objectives
- Focus of Agile Project Manager varies from phase to phase



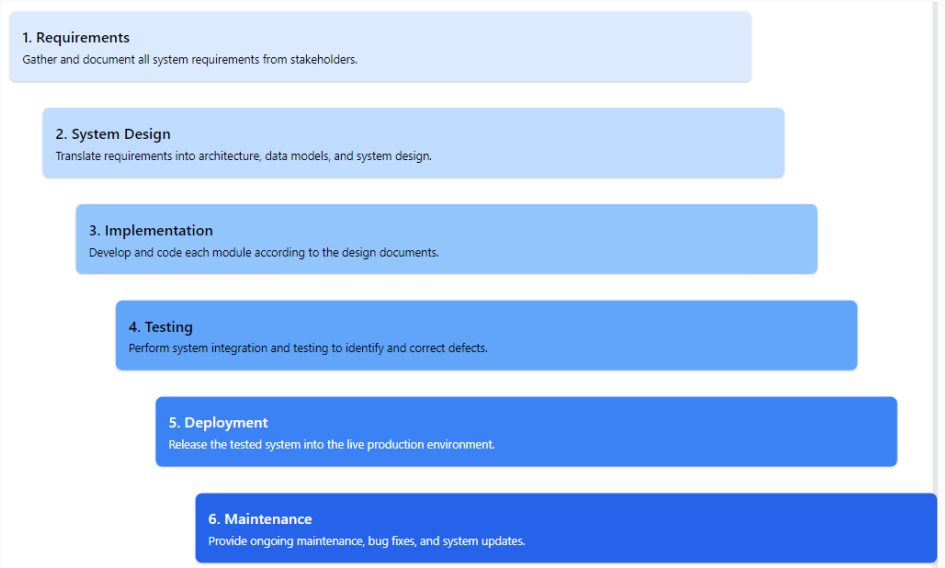
Overview : The DSDM Process

DSDM Phase	Purpose / Summary	Key Activities (Examples)	Key Deliverables (Examples)
1. Pre-Project	Ensures that only the *right projects* are started and that each is set up correctly based on clearly defined objectives.	• Identify potential projects at portfolio/program level • Confirm business justification for Feasibility • Allocate resources for next phase	• Project mandate or brief • Initial outline business case • Approval to proceed to Feasibility
2. Feasibility Phase	Determines whether the proposed project is viable—both technically and from a business perspective.	• Conduct business and technical feasibility study • Define outline requirements and solution options • Estimate high-level costs and duration • Plan resources for Foundations phase	• Outline Business Case (initial viability) • Outline Requirements List • Feasibility Assessment Report • Initial Delivery Plan (rough schedule)
3. Foundations Phase	Establishes firm foundations for the project by agreeing on what will be delivered, how, and when. Reduces uncertainty before development begins.	• Refine and prioritize requirements • Define technical architecture and development approach • Prepare management and delivery plans • Agree on strategy for deployment and change	• Agreed Business Case • Prioritized Requirements List • Solution Architecture Definition • Development & Management Approach Definitions • Delivery Plan (baseline)
4. Evolutionary Development Phase	Builds the solution incrementally through iterative development and frequent stakeholder feedback. Focus on delivering value early.	• Plan and execute Timeboxes • Develop, test, and integrate functional components • Conduct reviews and retrospectives • Adjust plans based on progress and feedback	• Evolving Solution (working increments) • Timebox Plans and Review Records • Updated Delivery Plan • Tested components ready for deployment
5. Deployment Phase	Deploys the solution (or increment) into operational use and transitions it to live environment.	• Coordinate release activities • Validate and deploy the solution increment • Conduct user training and change management • Hold Increment Retrospective	• Deployed Solution (live use) • Release/Deployment Plan • Project Review Report • User documentation and training materials
6. Post-Project Phase	Reviews business benefits after deployment to ensure project outcomes meet expectations.	• Measure and assess business benefits • Update business case based on actual results • Capture lessons learned • Plan for future increments if needed	• Benefits Assessment Report • Post-Implementation Review • Lessons Learned Summary

DSDM Phases vs Waterfall

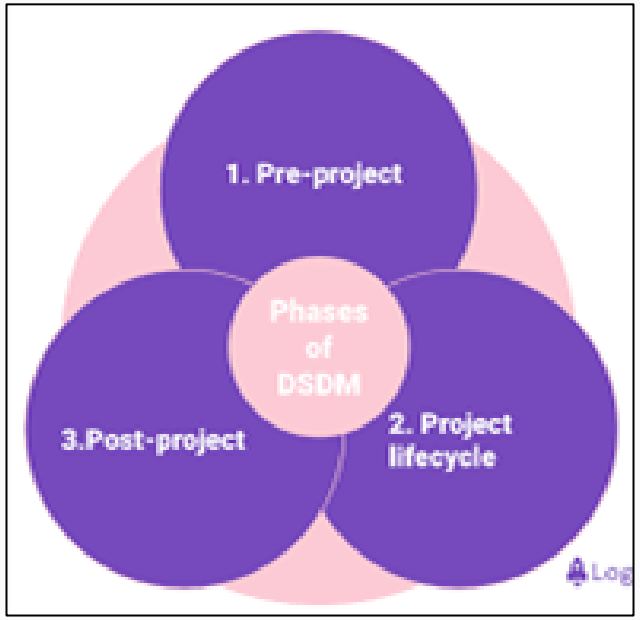
Comparison of DSDM and Waterfall Lifecycle Phases

DSDM Phase	Waterfall Equivalent Phase	Gap / Key Difference
1. Pre-Project	*Not explicitly defined*	Waterfall does not include a formal pre-project stage. DSDM ensures early justification, alignment with business strategy, and readiness before project initiation.
2. Feasibility Phase	Requirements Definition	DSDM validates technical and business feasibility before major investment. Waterfall assumes feasibility is already established and moves directly to detailed requirements gathering.
3. Foundations Phase	System Analysis / Design Planning	DSDM establishes a firm foundation — defining business goals, prioritizing requirements, and agreeing on architecture and delivery approach. Waterfall relies heavily on upfront documentation and rigid plans.
4. Evolutionary Development Phase	Design & Implementation	DSDM delivers iteratively in short, controlled Timeboxes with continuous stakeholder feedback. Waterfall follows a strict sequential flow from design to build without feedback loops.
5. Deployment Phase	Deployment / Implementation	Both deliver solutions into operational use, but DSDM emphasizes incremental releases, early value delivery, and active user involvement during deployment.
6. Post-Project Phase	Maintenance	DSDM includes formal review of business benefits and value realization. Waterfall focuses mainly on technical maintenance, bug fixes, and system updates after delivery.



DSDM Phase	Agile Project Manager Focus / Responsibilities
1. Pre-Project	• Minimal or no direct involvement (project not yet initiated). • May assist program/portfolio team in ensuring only the right projects are proposed. • Confirm resources are available for Feasibility phase.
2. Feasibility Phase	• Work with Business Analyst, Advisors, and Solution Development Team (SDT) members. • Evaluate project viability (business and technical). • Coordinate creation of outline plans and resources for Foundations. • Facilitate decision whether to proceed or stop the project.
3. Foundations Phase	• Collaborate with Business Analyst, Advisors, and SDT members to establish firm foundations. • Oversee preparation of key foundational products (Business Case, Requirements List, Architecture Definition, Delivery Plan). • Ensure all plans align with business objectives and timelines. • Baseline the Delivery Plan and commit to first Increment schedule.
4. Evolutionary Development Phase	• Delegate day-to-day management of Timeboxes to the SDT and Team Leader. • Protect the team from external distractions and ensure optimal environment. • Ensure resources and tools are available as planned. • Review Timebox progress and manage by exception. • Support Visionary and Technical Coordinator in planning Deployment.
5. Deployment Phase	• Coordinate day-to-day deployment activities. • Ensure readiness of both technical and business environments for live release. • Engage Business Visionary (for business change) and Technical Coordinator (for technical transition). • Facilitate Increment Retrospective and capture learning for future releases.
6. Post-Project Phase	• Typically no active involvement (project formally closed). • Ensure proper handover to Business Visionary for benefits realization. • Support lessons learned and post-project review if required.

Focus of Agile Project Manager varies from phase to phase



Overview : Planning in a DSDM Project

- Strategic planning starts in Feasibility

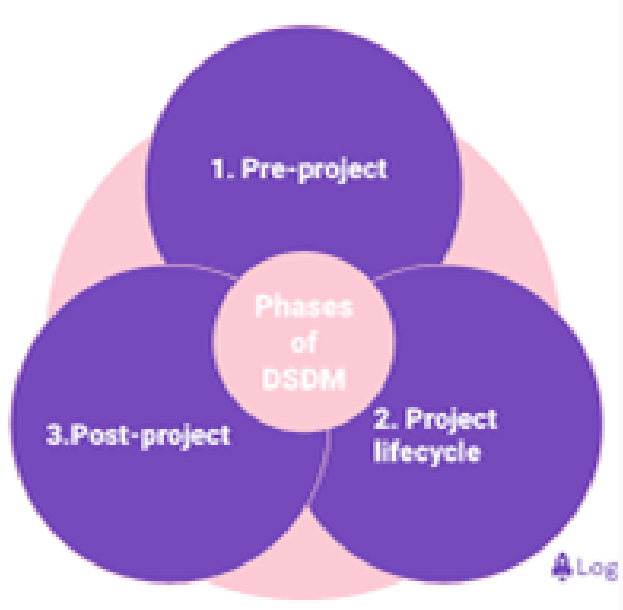
  - Initial outline of project management and solution development approaches

  - Supported by possible timeline for delivery

- This information gradually refined during Foundations

  - Generates high-level plans and agreed ways of working

  - Forms basis of commitment to deliver what business needs when business needs it

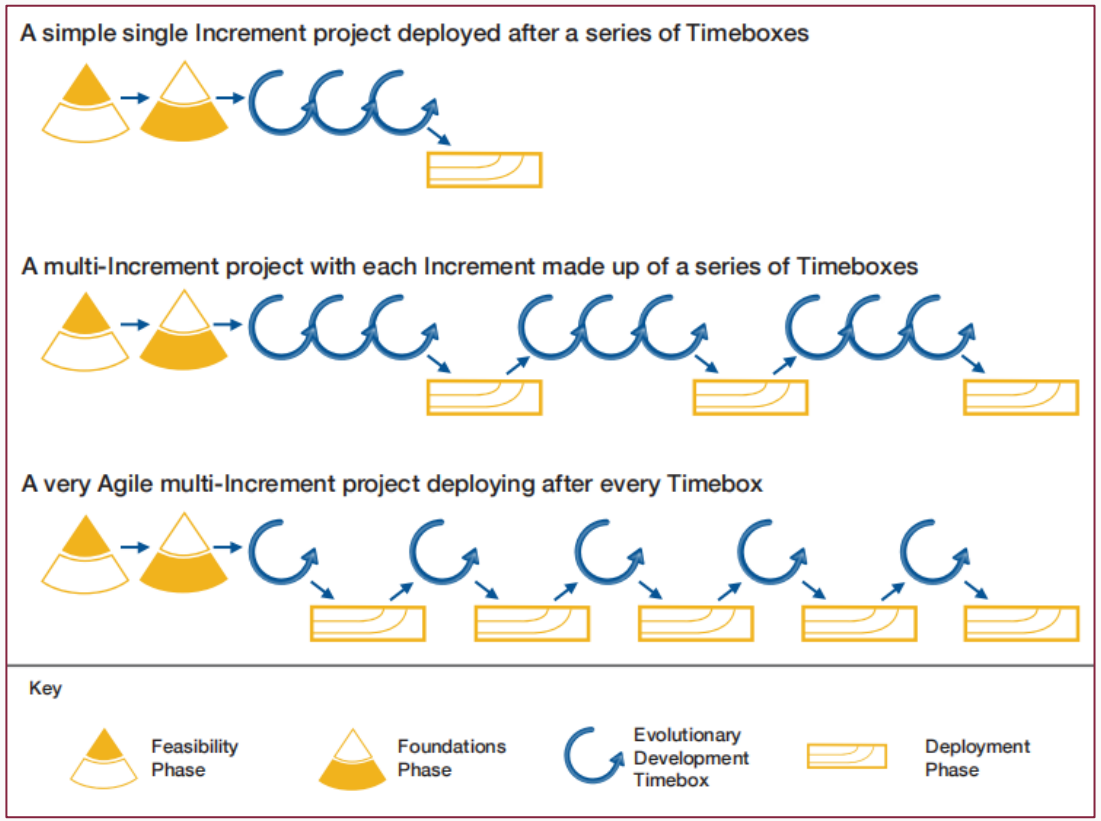


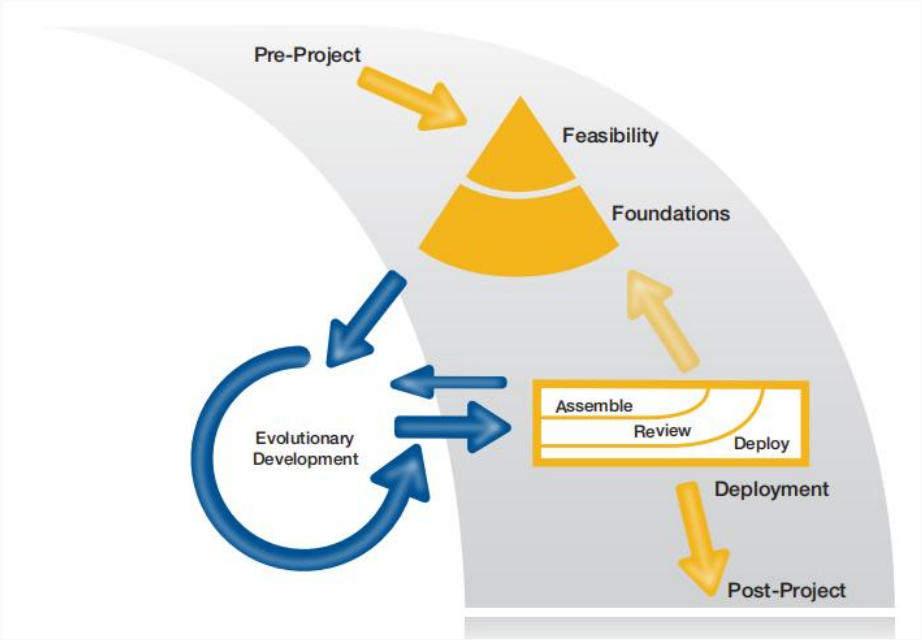
Overview : Planning in a DSDM Project

- Plans always need to evolve to meet changes in real-world circumstances
  - As business needs shift
  - As requirements and solution detail emerges
  - As understanding of what is possible and what is necessary changes
- Agile PM not responsible for all planning activity
- Key Agile PM responsibility to ensure all plans align with high-level plans to deliver what business needs, at time agreed, within agreed costs

Configuring the Process - 3 lifecycle examples

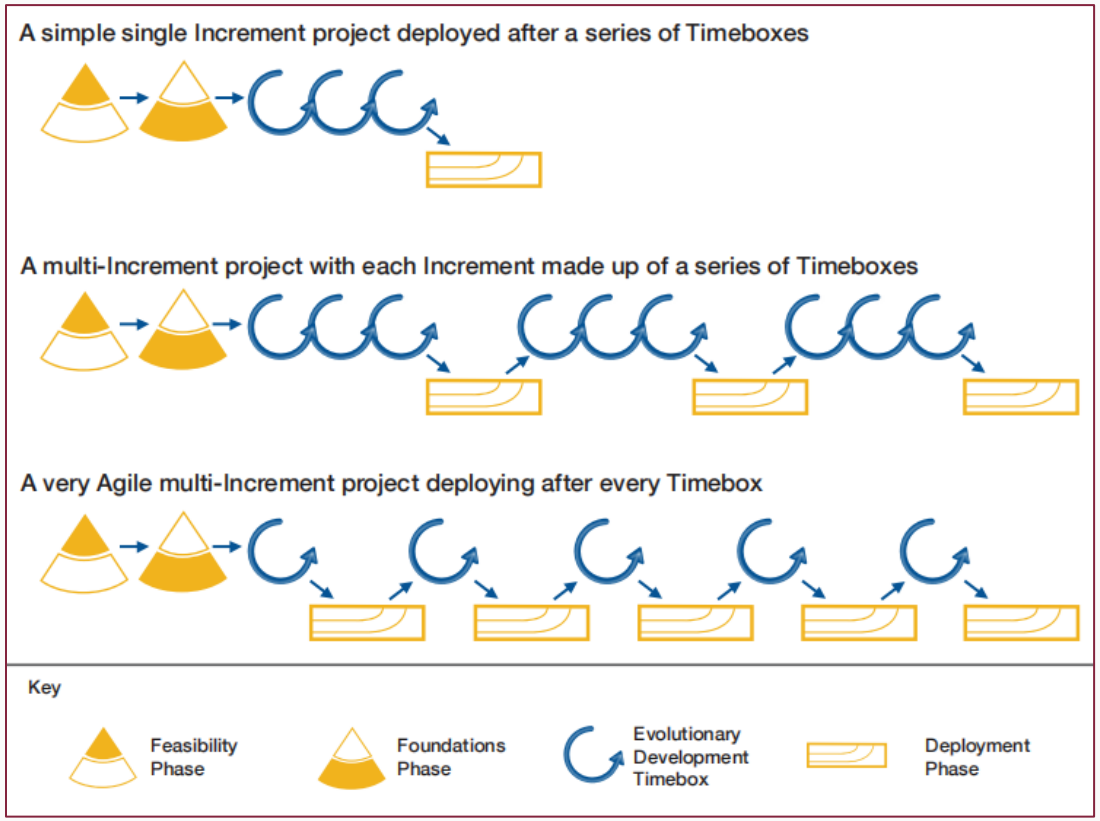
- Solution built and deployed incrementally from a firm foundation
- Ensure value delivered early and delivered frequently



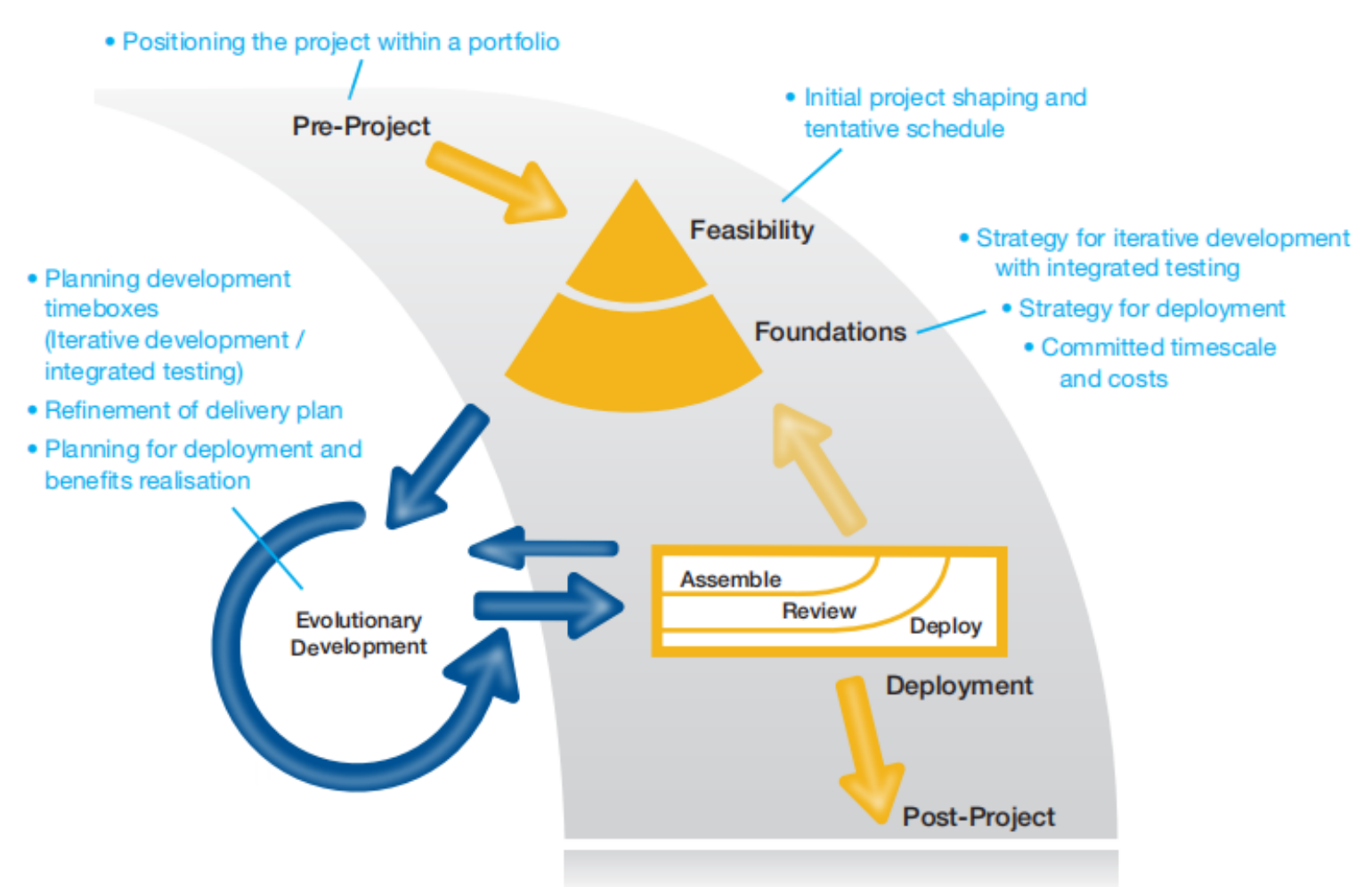


3 Options lifecycle

DSDM Lifecycle Option	Description / Implementation Approach	Example of How Phases Are Organized
1. Single-Increment Project	• The simplest form of DSDM implementation.• The project goes through Feasibility and Foundations once, then proceeds through several Timeboxes (iterations) in the Evolutionary Development Phase before a single Deployment at the end.• Suitable for smaller, well-defined projects with stable requirements.	**Sequence:**Feasibility → Foundations → (Multiple Evolutionary Development Timeboxes) → Single Deployment → Post-Project.**Example:** A small internal HR application built and deployed once after full iteration cycles.
2. Multi-Increment Project	• The project is divided into multiple Increments, each made up of several Timeboxes.• Each Increment represents a partial but usable version of the overall solution.• After each Increment, a Deployment occurs, followed by the next Increment cycle.• Suitable for medium-to-large projects where early releases deliver business value progressively.	**Sequence:**Feasibility → Foundations → Increment 1 (several Timeboxes → Deployment) → Increment 2 (several Timeboxes → Deployment) → Increment 3 (several Timeboxes → Deployment) → Post-Project.**Example:** An e-learning platform where each Increment adds a new module (e.g., user management, course upload, quiz engine).
3. Very Agile Multi-Increment Project	• The most agile configuration of DSDM.• Deployment occurs after every Timebox instead of after a series of Timeboxes.• Ensures very frequent delivery, tight feedback loops, and rapid adaptation to change.• Suitable for dynamic environments or continuous delivery setups.	**Sequence:**Feasibility → Foundations → (Timebox → Deployment → Timebox → Deployment ...) → Post-Project.**Example:** A SaaS web application that releases updates or features after every sprint (e.g., new dashboard, bug fixes, small enhancements).

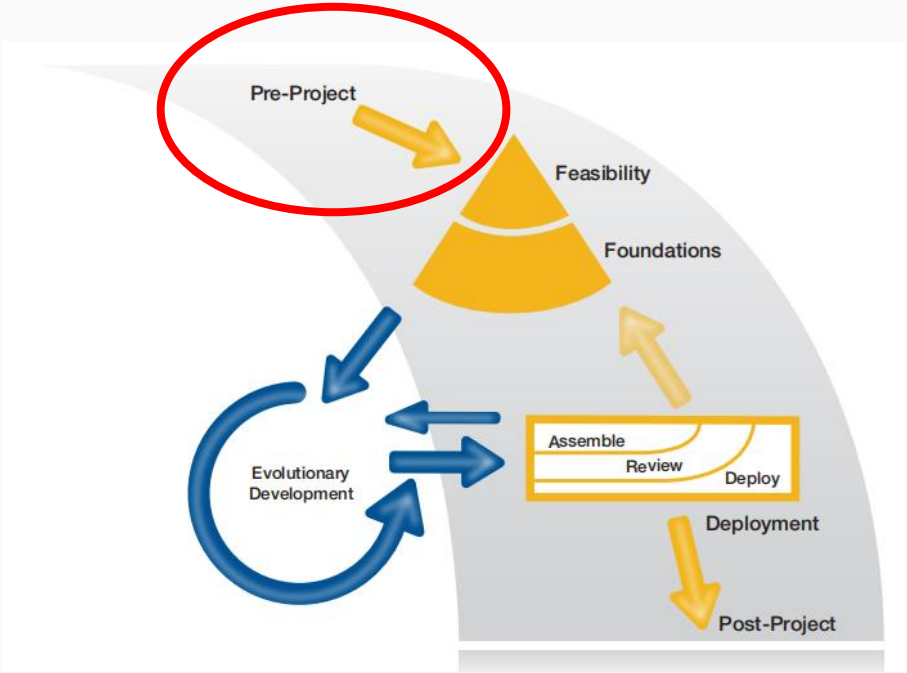


Planning Activity Phase by Phase



Pre-Project Phase

- Pre-Project
  - Ensure only the right projects are started
  - Ensure projects are set up correctly based on clearly defined objective
- Keep Pre-Project short and sharp
- Main purpose is to position project and justify Feasibility
- Formality varies, depending on organisation



There is no PM involvement Pre-Project

Pre-Project Phase (cont.)

- Planning at Pre-Project
  - Working at programme / portfolio level
  - Focused on whether and when Feasibility will be assessed
  - Ensuring Feasibility resources are available to do required work
  - Planning for detail of next phase (Feasibility)

Pre-Project

Feasibility Phase

- PM works with Project-level roles including

  Business Analyst PM may also include

  - Advisors (Business and Technical)

  - As many SDT members as possible

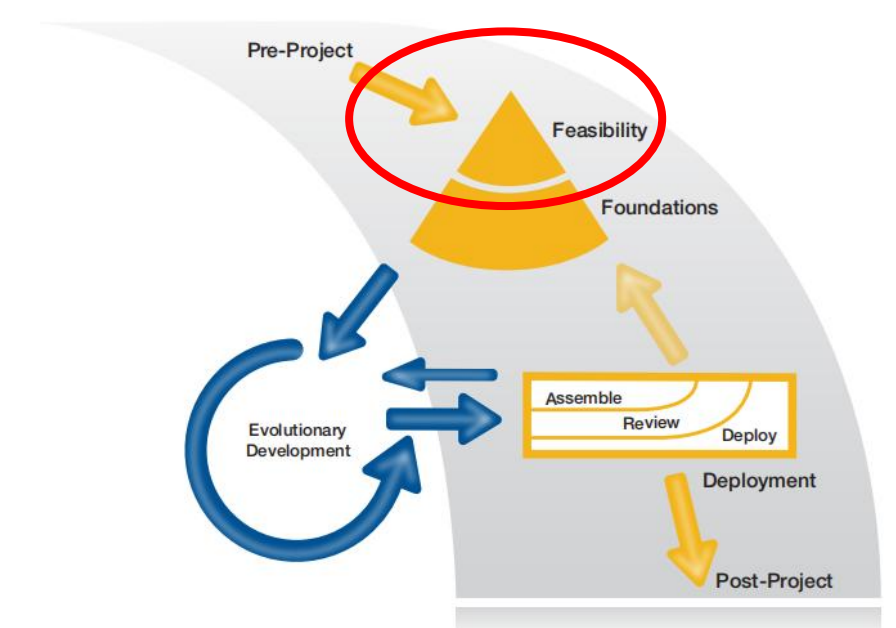
- Focus is to establish whether proposed project

  - Is likely to be feasible

    - Both from business and technical perspective

  - Would be cost-effective

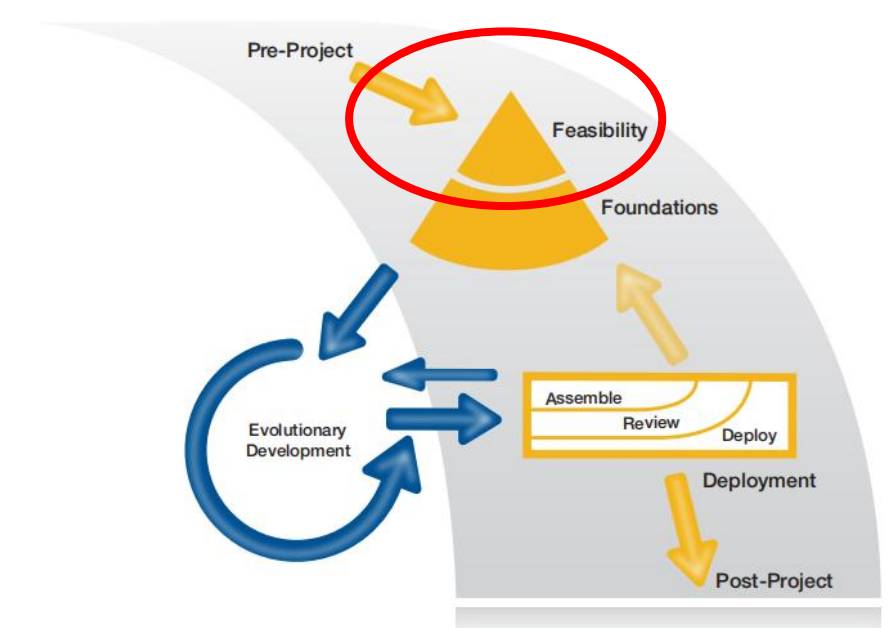
    - From business perspective



Feasibility Phase (cont.)

- To plan and resource Foundations phase

  - The six key Foundation products created in outline here
- Feasibility effort - just enough to decide whether further investigation is justified or project should be stopped if unlikely to be viable



Feasibility Phase (cont.)

- Planning during Feasibility
  - Based on outline information (requirements and solution)
  - Planning for detail of next phase (Foundations)
  - Providing an approximation of size and duration of overall project
- Based on what is known at this point - an educated guess
  - Delivery Plan describes
- Next few weeks in detail (Foundations)
- High-level outline for first Increment
- Possibly list of proposed dates for Increments into the future
  - Plan includes timescale, deliverables, resources and facilities needed
  - Not yet possible to plan any detail for Evolutionary Development or Deployment

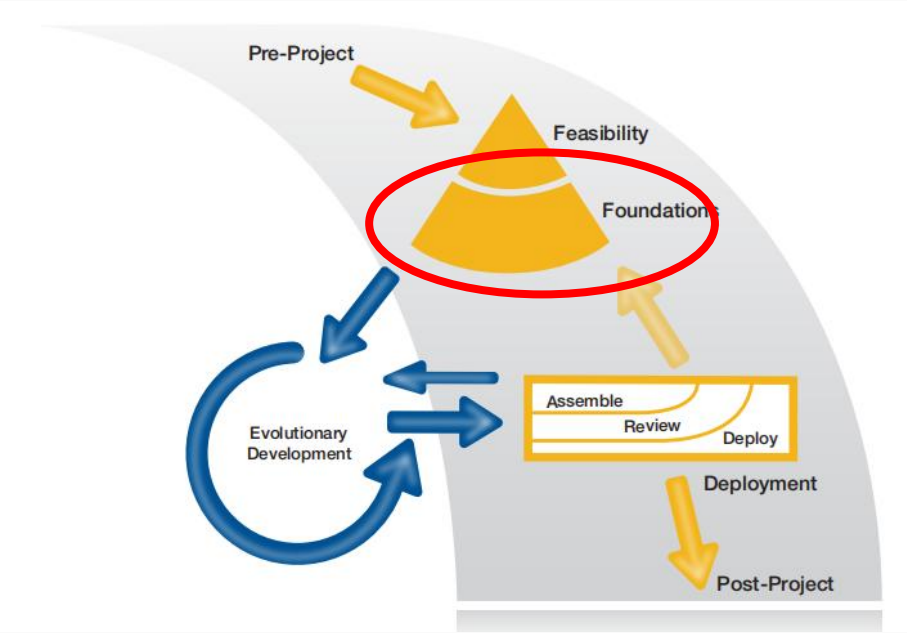
Pre-Project

Feasibility



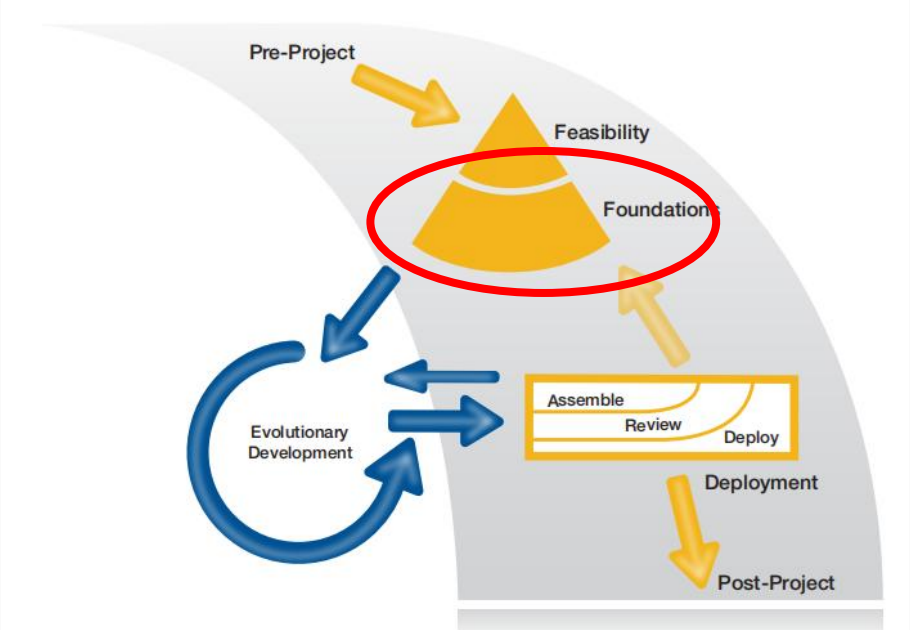
Foundations Phase

- PM works with Project-level roles, including Business Analyst and involving:
  - Business and Technical Advisors
  - As many SDT members as possible
- Organising work to establish firm Foundations
  - From Business perspective
    - Agreed realistic Business Case (Recommended)
    - Prioritised Requirements List (Mandatory)



Foundations Phase (cont.)

- From Solution / Technical perspective
    - Solution Architecture Definition and Development Approach Definition
- From Management perspective
    - Delivery Plan (Mandatory)
    - Management Approach Definition
    - Foundations Summary
- PM establishes which of 6 key Foundations products need to be created or evolved from Feasibility outline set



Foundations Phase (cont.)

- Based on
  - More detailed understanding of requirements and priorities
  - Reduced uncertainty and increased accuracy of estimates
- Focus is on
  - Creating schedule of Timeboxes for first Increment, with resources needed
  - Defining approaches to be used for developing and controlling development of solution
  - Agreeing strategy for Deployment and assessing high-level impact of change and training needs
    - Delivery Plan baselined at end of Foundations
    - Delivery date for first Increment is committed
- Delivery Plan also provides
  - Probable focus for each Timebox in this Increment
  - Resources required to evolve solution
- Delivery Plan does not include detailed Timebox objectives and tasks

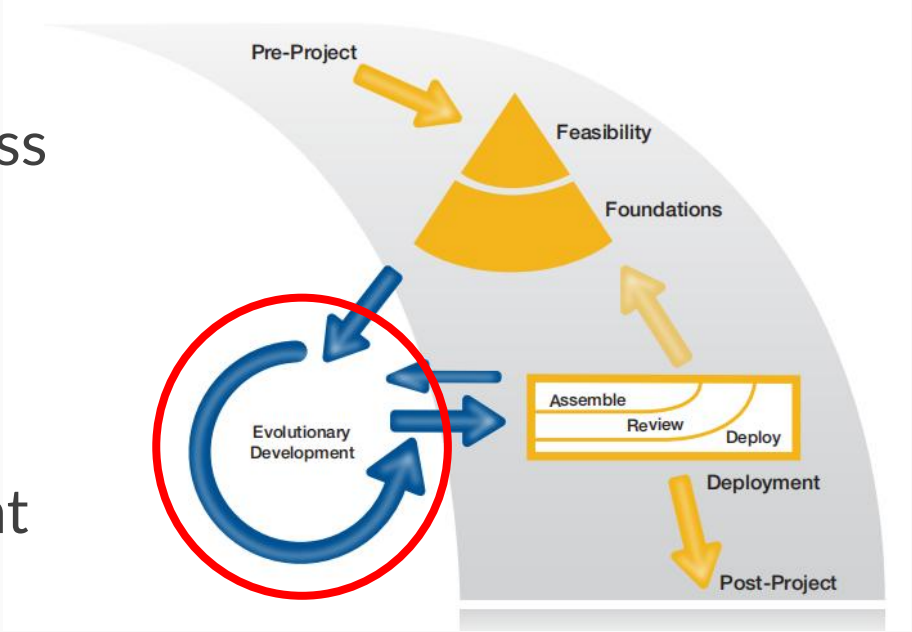
Foundations



Evolutionary Development Phase

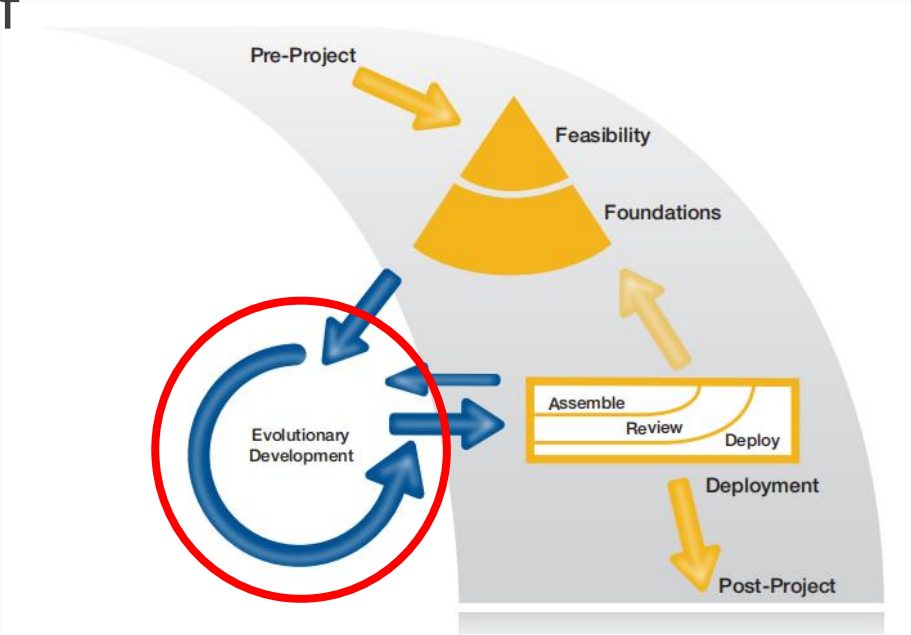
- Evolutionary Development builds on project firm foundations to evolve the solution
  - Through use of DSDM Practices
  - Converging on accurate solution that meets business need and technically is built the right way

- PM passes day-to-day management of Timeboxes to SDT
  - PM focuses mainly on ensuring project environment optimised for agile development
  - Protecting ST from distractions
  - Ensuring tools and facilities available as planned



Evolutionary Development Phase (cont.)

- During Evolutionary Development, PM
  - Participates in (and may facilitate) Timebox Kick-off
  - Participates in Timebox Close-out
    - To measure progress, to feed learning and incomplete work back into Delivery Plan
  - Ensures people and resources are committed for duration of Timebox
  - Manages engagement of Business Ambassadors, Advisors and any part-time resources
  - "Manages by exception"
- PM works primarily with Visionary and Technical Coordinator to plan Deployment



Evolutionary Development Phase (cont.)

- Timebox planning at beginning of each Timebox

  - This is lowest level of DSDM Planning

  - SDT members responsible for Timebox planning

- Timebox Plans based on

  - Objectives and outcomes plus Timebox length agreed at Timebox Kick-off

  - Task-level estimates that emerge as result of detailed investigation of requirements

  - Decisions made on how requirements will be fulfilled



Evolutionary Development Phase (cont.)

- Timebox Plans

  - Typically captured on Team Board (or electronically)

  - Indicate who will do what work to achieve Timebox objectives and generate agreed outcome

- Team Leader ensures

  - All work is covered by the plan

  - Resources are sufficient to do majority of work agreed

  - Flagging any significant issues that may arise as TB progresses



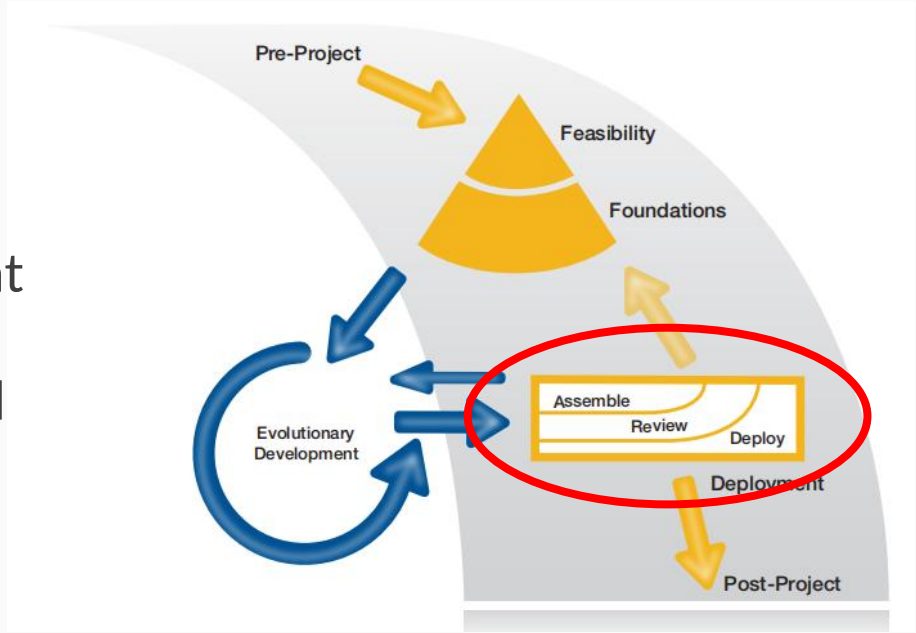
Deployment Phase

Deployment

- To bring a baseline of Evolving Solution into operational use
  - This may be the final solution or a subset of final solution
- After last release, Project is formally closed

Project Manager

- Coordinates day-to-day activities to bring Solution Increment into live operational use
- Ensures everything is coherent and right people are involved
  - Business Visionary (responsible for business change)
  - Technical Coordinator (responsible for technical transition into live use)
  - Collaborating with ST and other stakeholders
- Organises Increment Retrospective
  - Results captured in Project Review Report



Deployment Phase (cont.)

- Planning for Deployment

  - As detail emerges during Evolutionary Development, plans for Deployment can be considered in more detail

  - Deployment involves everything needed to transition solution into live use

  - Delivery Plan is updated with Deployment activities as they become clear

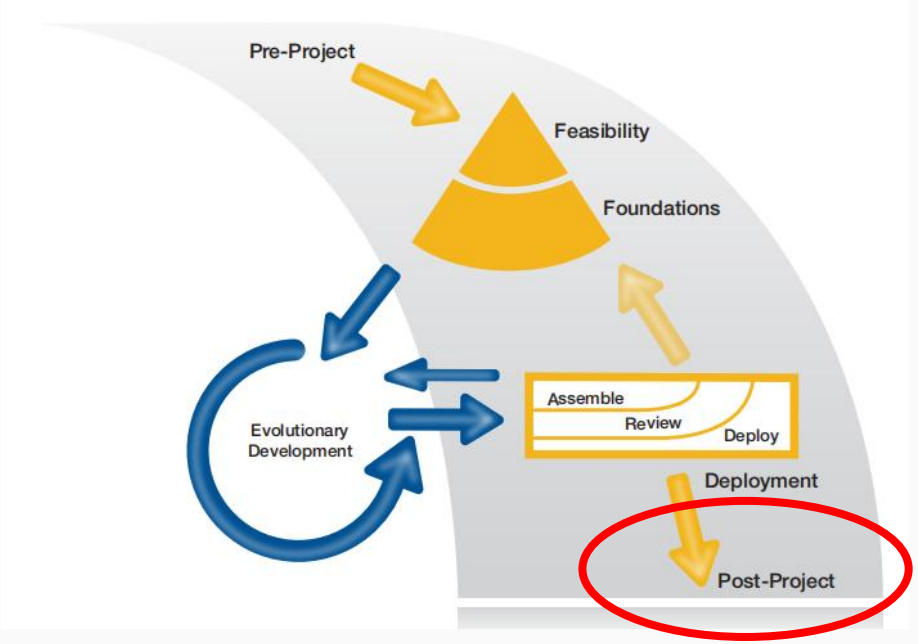
  - Plans for Deployment should always include roll-back options



Post-Project Phase

Post-Project

- To check how well expected business benefits have been met
- There is no PM involvement here as the project has been closed down by this point
- Planning benefits assessment activity, Post-Project
  - Based on clearer Deployment plans



Post-Project Phase (cont.)

- Activities needed Post-Project to measure benefits can now be planned
- Activity for this owned by Business Visionary, who owns wider implications change
- Incremental nature of DSDM means revisiting Foundations at end of each Increment helps :
  - Check project as whole remains viable and should proceed to next Increment
- Based on validated actual effort and updated more informed view of Business Case
  - Firm up plans for the subsequent Increment

Post-Project

Planning - Agile PM Top Tips

- Remember DSDM planning activities are always highly collaborative.

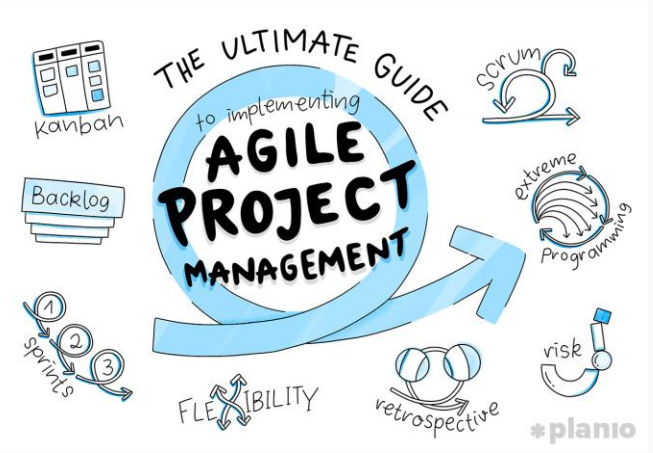
  - Collaboration brings in expertise of those executing the plans

  - Collaboration improves buy-in to the plans

- This is especially important for Delivery Plan
- Timebox Planning is not your responsibility - it is owned by SDT

  - You only need to review Timebox Plan

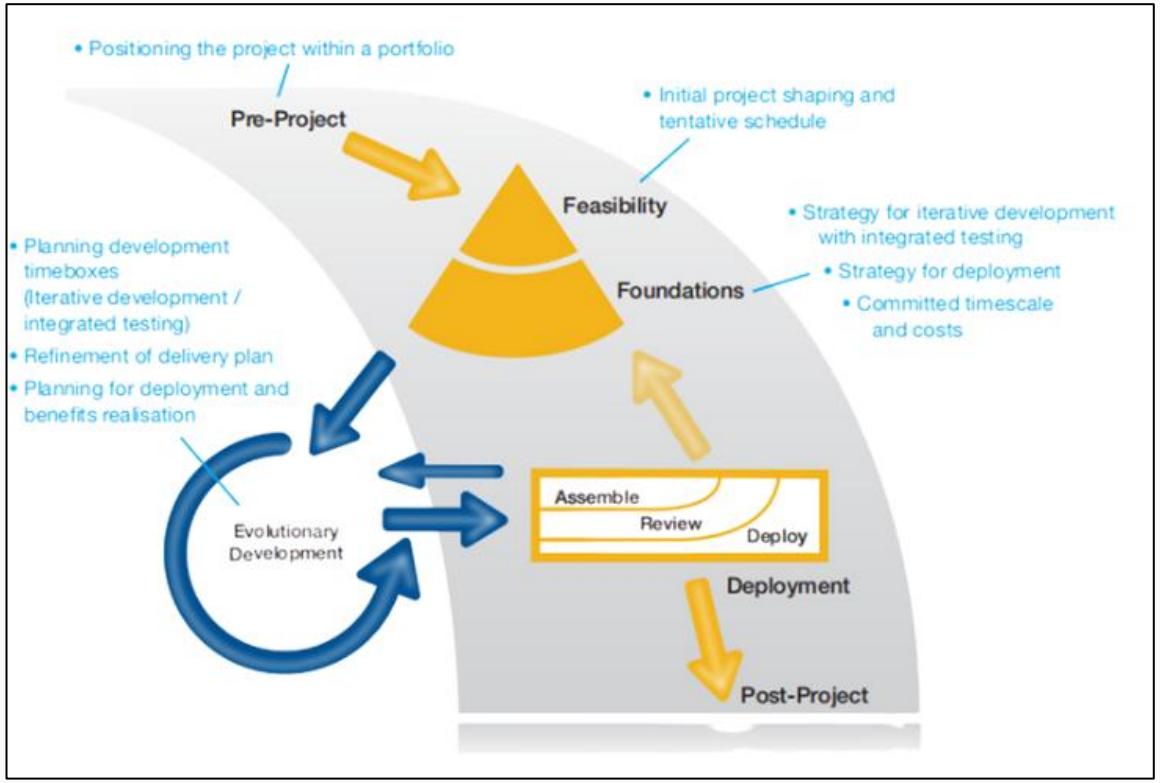
    - To ensure it aligns with agreed TB objectives and that it remains aligned



- Any tools supporting planning activity should not work against or constrain effective DSDM planning
- When planning, always check to ensure work identified adds real value to the project

Agile PM – Planning and Monitoring

DSDM Phase	What Needs to Be Planned by the Agile Project Manager (PM)	What Needs to Be Monitored by the PM
1. Pre-Project	• Ensure project initiation readiness and justification.• Plan resource availability for Feasibility phase.• Outline the initial timeline for project assessment.	• Confirm that Feasibility preparation tasks are completed on time.• Monitor alignment with program or portfolio objectives.
2. Feasibility Phase	• Develop a high-level Delivery Plan for Foundations.• Plan required resources, roles, and time for feasibility analysis.• Identify key risks and constraints early.	• Track progress of feasibility study and data collection.• Monitor decision-making for project continuation or termination.• Ensure feasibility outcomes are well-documented.
3. Foundations Phase	• Plan overall project structure, delivery approach, and schedule.• Define Timebox schedule for the first Increment.• Plan management and communication approaches.• Establish Delivery Plan baseline.	• Monitor creation of the six key Foundation products.• Review plan alignment with business case and priorities.• Ensure Delivery Plan is realistic and agreed upon by stakeholders.
4. Evolutionary Development Phase	• Plan resource allocation across Timeboxes.• Ensure environment and tools are ready for development.• Plan Deployment strategy and high-level change management.	• Monitor Timebox progress via reviews and retrospectives.• Review Timebox Plans (created by SDT) to ensure alignment with objectives.• Manage risks, dependencies, and exceptions.
5. Deployment Phase	• Plan deployment schedule and transition activities.• Coordinate resource readiness for release and support.• Plan Increment Retrospective and knowledge transfer.	• Monitor deployment progress and issue resolution.• Ensure business and technical teams collaborate effectively.• Review feedback from deployment and lessons learned.
6. Post-Project Phase	• Plan benefits realization assessment.• Schedule post-implementation review.	• Monitor collection of benefit data and performance metrics.• Ensure lessons learned are captured and communicated for future projects.



Configuring DSDM for Scalability and Formality

DSDM recognises the value of Agility:

- Improves project productivity.
- Enhances solution quality.
- Accepts constraints in corporate environments.

Common corporate constraints:

- Financial governance.
- Architecture and/or infrastructure strategies.
- Regulatory governance.
- Vendor agreements.
- Third-party support considerations.



Configuring DSDM for Scalability and Formality

Configurability of the DSDM process:

- Can be tailored for different project sizes.
- Small projects → lighter governance.
- Large projects → stronger governance.
- Lifecycle configured to suit the project.
- Formality of DSDM products defined, created, and approved at appropriate levels.

Scaling project organisation:

- Structure can be refined to support multiple teams.
- Key roles act as directors and coordinators across teams.
- For complex projects, products can be made more detailed and formal:
  - Solution Architecture Definition.
  - Development Approach Definition.
  - Management Approach Definition.
  - Delivery Plan and Timebox Review Records.

🧩 Exercise #4 – Analyzing the DSDM Lifecycle for HIP 2.0 Project

Instruction:

Based on the HDC's Halal Integrated Platform (HIP 2.0) case study and the DSDM Agile Project Framework lifecycle, analyze which DSDM lifecycle option (Single Increment, Multi-Increment, or Very Agile Multi-Increment) is the most suitable for this project.

You are required to:

- Analyze and justify your choice of lifecycle option based on the project's characteristics, deliverables, and release plan (refer to the project charter).
- Compare the advantages and limitations of each option before concluding which one best fits the HIP 2.0 implementation.
- Draw a labeled diagram to illustrate your proposed solution — showing the flow of DSDM phases (Pre-Project → Feasibility → Foundations → Evolutionary Development → Deployment → Post-Project) and how each release or increment is delivered within the lifecycle.
- Include a short explanation (150–200 words) to support your diagram, describing how your chosen lifecycle option ensures timely delivery, quality control, and stakeholder involvement.



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